CSC 144V: Autonomous Vehicles
Lecture 5

Object Detection and Segmentation Exercises Solution

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Exercise: IoU and NMS

Four detections of two pedestrians, boxes (x_1, y_1, x_2, y_2) in pixels. IoU = $|A \cap B| / |A \cup B|$.

(a) Compute the IoU of each box with the top-scoring box A. (b) Run greedy NMS: keep the best box, suppress boxes with IoU > threshold with a kept box. Threshold 0.5. (c) Which boxes remain with threshold 0.4?

Box	(x_1, y_1, x_2, y_2)	Score	IoU with A	NMS, thr. 0.5
A	(10, 10, 60, 60)	0.9		
B	(20, 20, 70, 70)	0.8		
C	(12, 12, 62, 62)	0.7		
D	(100, 100, 150, 160)	0.6		

Exercise Solution: IoU and NMS

- IoU(A, B): intersection 40 × 40 = 1,600, union 2,500 + 2,500 − 1,600 = 3,400 → **0.47**. IoU(A, C): 48 × 48 = 2,304, union 2,696 → **0.85**. IoU(A, D) = **0**.
- Threshold 0.5: keep A; suppress C (0.85); B survives (0.47 ≤ 0.5); D survives → **A, B, D**.
- Threshold 0.4: B is also suppressed (0.47 > 0.4) → **A, D**. A low threshold risks removing a second object that overlaps the first.

Box	(x_1, y_1, x_2, y_2)	Score	IoU with A	NMS, thr. 0.5	NMS, thr. 0.4
A	(10, 10, 60, 60)	0.9	1	keep	keep
B	(20, 20, 70, 70)	0.8	0.47	keep	suppress
C	(12, 12, 62, 62)	0.7	0.85	suppress	suppress
D	(100, 100, 150, 160)	0.6	0	keep	keep

Calculating IoU(A, B)

- The coordinates of the boxes are (lower left corner and upper right corner):
  - Box A: (10,10,60,60)
  - Box B: (20,20,70,70)
- Each box has a width and height of 50, so each has an area of:
  - Area(A)=50×50=2500
  - Area(B)=50×50=2500
- **Step 1: Calculate the intersection.**
  - The overlapping region extends from coordinate 20 to 60 on both axes, with width and height both 40
  - Intersection=40×40=1600
- **Step 2: Calculate the union.**
  - Subtract the overlapping area from the sum of both box areas:
  - Union=2500+2500−1600=3400
- **Step 3: Calculate IoU.**
  - IoU(A,B)=1600/3400≈0.47
- This means the boxes have approximately **47% overlap** relative to their combined area.

Next Slides are not covered in Midterm

Exercise: Precision, recall and AP

There are **5** ground-truth objects. The detector returns 6 detections sorted by confidence: TP, TP, FP, TP, FP, TP.

(a) Precision and recall after each rank.

(b) Average precision using the all-point interpolated envelope: precision at recall r is the maximum precision at any recall ≥ r.

Rank	TP / FP	Precision	Recall
1	TP		
2	TP		
3	FP		
4	TP		
5	FP		
6	TP		

Exercise Solution: Precision, recall and AP

- Precision = TP/rank; recall = TP/5.
- Envelope: recall 0.2 → 1, 0.4 → 1, 0.6 → 0.75, 0.8 → 0.667, 1.0 → 0 (never reached).
- AP = 0.2 × (1 + 1 + 0.75 + 0.667 + 0) ≈ **0.683**.

Rank	TP / FP	Precision	Recall
1	TP	1.00	0.2
2	TP	1.00	0.4
3	FP	0.67	0.4
4	TP	0.75	0.6
5	FP	0.60	0.6
6	TP	0.67	0.8

Exercise: YOLO label of a grid cell

A **300 × 300** image is divided into a **3 × 3** grid; classes: Pedestrian, Car, Motorcycle. A car has its mid-point at pixel (170, 240); its box is **150 px wide** and **60 px high**.

(a) Which grid cell (row, col from 0, top-left) is responsible?

(b) Fill in the label y = (pc, bx, by, bh, bw, c_1, c_2, c_3) of that cell.

(c) What is the shape of the whole output tensor?

pc	bx	by	bh	bw	c_1	c_2	c_3

Exercise Solution: YOLO label of a grid cell

- (a) Cell size 100 px: col = ⌊170/100⌋ = 1, row = ⌊240/100⌋ = 2 → **cell (2, 1)**.
- (b) bx = (170 − 100)/100 = 0.7, by = (240 − 200)/100 = 0.4, bh = 60/100 = 0.6, bw = 150/100 = 1.5 (can exceed 1). Car is class 2.
- (c) 8 numbers per cell: **3 × 3 × 8**.

pc	bx	by	bh	bw	c_1	c_2	c_3
1	0.7	0.4	0.6	1.5	0	1	0

Exercise: Realistic YOLO dimensions

A realistic YOLO takes a **608 × 608 × 3** image and outputs a **19 × 19 × 5 × 85** tensor.

(a) What do 5 and 85 stand for?

(b) How many numbers are predicted, and how many candidate boxes enter NMS?

(c) What is the stride (pixels per grid cell)? In which cell (row, col from 0) does an object with mid-point at pixel (330, 200) fall, and what are (bx, by) of its label?

Exercise Solution: Realistic YOLO dimensions

- (a) **5 anchor boxes** per cell; 85 = 5 + 80: (pc, bx, by, bh, bw) plus **80 class scores**.
- (b) 19 × 19 × 5 × 85 = **153,425** numbers; 19 × 19 × 5 = **1,805** candidate boxes before NMS.
- (c) Stride 608/19 = **32 px**. col = ⌊330/32⌋ = 10, row = ⌊200/32⌋ = 6 → **cell (6, 10)**; bx = (330 − 320)/32 = **0.31**, by = (200 − 192)/32 = **0.25**.

Exercise: Upsampling

Upsample the 2 × 2 feature map to 4 × 4 with a stride of 2.

(a) **Bed of nails**: insert zeros; each input goes to the top-left corner of its 2 × 2 block.

(b) **Nearest neighbor**: duplicate each element.

(c) Name two other upsampling methods used in FCNs.

Input (2 × 2)

1	2
3	4

Bed of nails (4 × 4)

Nearest neighbor (4 × 4)

Exercise Solution: Upsampling

- (a) Bed of nails keeps the values at fixed positions and fills the rest with 0.
- (b) Nearest neighbor replicates each value over its 2 × 2 block.
- (c) Max unpooling (restores each value to the position of the max in the matching pooling layer), bilinear / bicubic interpolation, transposed convolution (learnable).

Input (2 × 2)

1	2
3	4

Bed of nails (4 × 4)

1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

Nearest neighbor (4 × 4)

1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

Exercise: Transposed convolution

Transposed convolution (stride 1, no padding): each input element multiplies the whole kernel, the products are placed at its position, and overlapping outputs add up.

(a) Compute the 3 × 3 output.

(b) Output size formula for input N, kernel K, stride S?

(c) With L stacked 3 × 3 CONV layers (stride 1) the receptive field is 2L + 1. For L = 10? How many layers for a receptive field ≥ 224? How does downsampling help?

Input

1	2
3	4

Kernel

1	2
0	1

Output (3 × 3)

Exercise Solution: Transposed convolution

- (a) e.g. O[1][1] = 1·1 + 2·0 + 3·2 + 4·1 = 11 (four overlapping contributions).
- (b) (N − 1)·S + K = (2 − 1)·1 + 2 = **3**.
- (c) L = 10 → **21 × 21**. 2L + 1 ≥ 224 → L ≥ 111.5 → **112 layers**: too slow. Pooling / strided convolution makes the receptive field grow much faster and shrinks the cost; upsampling then restores the output size.

Input

1	2
3	4

Kernel

1	2
0	1

Output (3 × 3)

1	**4**	**4**
3	**11**	**10**
0	**3**	**4**